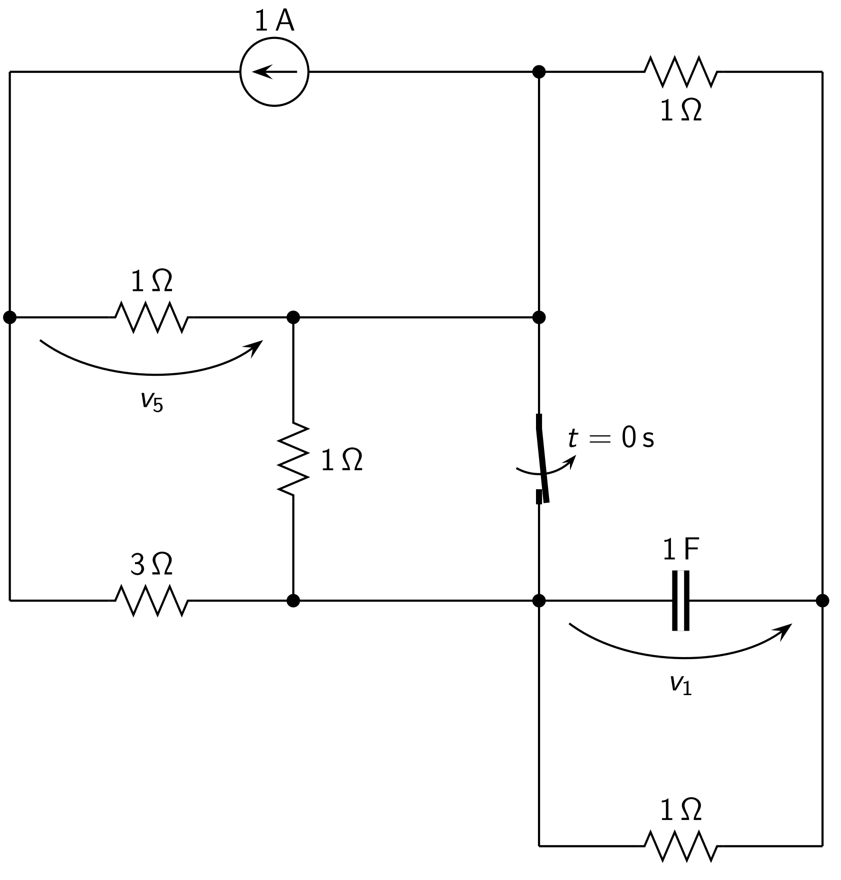
Problem: find the expression of $v_1(t)$, $v_5(t)$ for $t > 0$.



Solution

- Time constant: $\tau = \frac{9}{14}\,\mathsf{s}$;
- Initial conditions before switching: $v_1(0^-) = 0\,\mathrm{V}$;
- Initial conditions after switching: $v_1(0^+) = 0\,\mathrm{V}$; $v_5(0^+) = \frac{-7}{9}\,\mathrm{V}$;
- Steady-state solution: $v_1(\infty) = \frac{-1}{14}\,\mathrm{V}$; $v_5(\infty) = \frac{-11}{14}\,\mathrm{V}$;
- Solution for $t > 0$:

$$v_1(t) = \left(\frac{1}{14}\,e^{-t/\tau} + \frac{-1}{14}\right)\,\mathrm{V}$$

$$v_5(t) = \left(7.937\cdot 10^{-3}\,e^{-t/\tau} + \frac{-11}{14}\right)\,\mathrm{V}$$